

Joshua A. Duarte

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Education

University of California, Berkeley , Berkeley, California Master of Engineering in Mechanical Engineering	Graduation: May 2023 GPA: 3.861/4.0
Texas A&M University , College Station, Texas Bachelor of Science in Mechanical Engineering	Graduation: May 2022 GPA: 3.594/4.0

Professional Experience

Lucid Motors , Newark, California <u>Senior Mechanical Engineer, Drive Unit</u>	April 2026 - Present
- Leading cross-functional design, integration, and validation efforts across the Lucid Air, Gravity, and Sapphire drive unit programs - Drive hardware changes from concept through CAD, prototype builds, testing, design iteration, and production release - Evaluate design options and communicate trade-offs across performance, reliability, cost, timing, and manufacturability	
<u>Mechanical Engineer II, Drive Unit</u>	January 2024 - April 2026
- Designed and validated components, interfaces, and assemblies for complex electromechanical drive unit systems - Performed CAD (CATIA V6), GD&T, tolerance stack-ups, bolted-joint analysis, DFM/DFA reviews, and drawing release activities - Built and evaluated trial hardware through fit checks, assembly testing, failure replication, and dimensional inspection - Resolved production and field issues with manufacturing, quality, test, service, supply chain, and supplier teams	
Boeing , Seattle, Washington <u>Boeing Test & Evaluation Flight Test Design/Build Intern</u>	May 2021 - August 2021
- Responsible for the design, prototypes, manufacturing, and installation of foreign object debris shields within 4 777X aircrafts - Worked closely with lead design engineers in overseeing the project management for the installation of essential test equipment designed to manipulate the center of gravity and electrical load throughout an aircraft during flight testing	
ExxonMobil , Houston, Texas <u>UIS Wells Digital Innovation & Automation Intern</u>	June 2020 - August 2020
- Developed and enhanced a SharePoint tool that captured and distributed learnings found during the Well construction process - Designed a user interface for data collection using Nintex Forms/Workflows, JavaScript, jQuery, HTML, along with VBA within Excel	
Northrop Grumman , Palmdale, California <u>Technical College Intern, Aerospace Systems, Top-Secret Clearance</u>	May 2019 - August 2019
- Assembled and tested electronic, mechanical, and electromechanical systems to ensure consistency and quality - Constructed models, assemblies, and engineering drawings using 3D modeling and design software, CATIA V5, to fabricate solutions	

Projects

Aucosto.com , Fremont, California - Building a full-stack personal operations platform unifying planning, time tracking, finance, habits, projects, and task management - Designed a modular dashboard with reusable widgets, shared event logging, secure data flows, and service-based architecture	February 2026 - Present
Blue Goji: Omni-Directional Treadmill , Berkeley, California - Improved the industrial design and overall efficiency of an Omni-Directional treadmill through DFM/DFA practices - Utilized fabrication techniques including injection, machining, and stamping to reduce overall footprint of the system by 57% - Created and managed comprehensive project plans, including Gantt Charts, budgets, BOMs, and resource allocation for a team of 10	August 2022 - May 2023
Gambit , College Station, Texas Persistently applied mechanical, electrical, and computer software knowledge to design and manufacture an intelligent chessboard capable of moving opponent chess pieces autonomously through a self-contained core XY gantry system	January 2021 - May 2023

Leadership Experience

Lucid Motors HOLA Employee Resource Group (ERG) <u>Vice President</u> - Oversee ERG operations, execution of initiatives, stakeholder engagement, and alignment with broader DEI and business objectives - Championed the development of the 2025-2026 strategic plan, including KPI definition, membership growth initiatives, engagement metrics, professional development opportunities, and long-term community impact goals	May 2025 - Present
Society of Hispanic Professional Engineers , College Station, Texas <u>President</u> - Led 8 board members to continuously impact members through professional, academic, technical, and social development which resulted in more than 400 members registered with the organization, a 74.7% increase in members from the previous year - Coordinated a 36.2% increase in corporate sponsorship and an 181% increase in event attendance from recorded metrics through new initiatives tailored for professional, academic, social, and technical development	May 2021 - May 2022

Skills

- CAD (CATIA V6/SolidWorks/Fusion 360), Programming & Data Analysis, DFM/DFA, FEA, GD&T, Bilingual (English/Spanish)